

Sina Ronaghi

Metrologist | Data Analyst | Researcher
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LinkedIn | ResearchGate | GitHub | Online Portfolio

PROFILE SUMMARY

Multidisciplinary Electrical Engineer (Ph.D.) specialized in **Metrology**, **Data Science**, and **Signal Processing**. Proven track record within the high-tech industrial and R&D settings with a particular focus on entrepreneurship and project management. **Experienced leader** in startups and EU Flagship Programs with a hands-on and engaging attitude. Driven by curiosity and passionate about adopting **emerging AI technologies** for process optimization, task coordination, and improving team efficiency.

CORE COMPETENCIES

Signal Processing	Waveform Analysis, Harmonic Characterization and THD Analysis, Digital & Statistical Signal Processing, Sensor Fusion.
Engineering Tools	Python, MATLAB & R (Pattern Recognition, Predictive Maintenance, Machine Learning, Data Visualization & Signal Processing), Databases (SQL, MongoDB, InfluxDB), Circuit Design & Analysis (Altium, LTSpice), Instrumentation & Automation (NI LabVIEW/TestStand)
Metrology Framework	JCGM/GUM, CENELEC, WELMEC, EN, IEC, MID
Regulatory Exposure	EN 50160 (Voltage Characteristics), IEC 62052/53/56 (Smart Metering), IEC 61000 (EMC), ISO/IEC 17025 (Testing, Inspection) & Certification, ISO 9001 (QMS)
AI & Automation	Vertex AI, n8n, Node-RED, Ollama, Notebook LM, AWS, AZURE & GCP
Soft Skills	Certified LEAN Manager (Kaizen, 5S), Agile & DevOps (Trello, Jira & GitLab), Proactive, Public speaking, Cross-disciplinary stakeholder engagement

PROFESSIONAL EXPERIENCE

Post-Doc Researcher / Project Manager **Nov 2025 – Present**

Politecnico di Milano *Milan, Italy*

- Optimized R&D testing frameworks for VT/CT characterization for LV/MV voltage characteristics (EN 50160) for Smart Grid Optimization in MATLAB/Python, increasing testing throughput by 70%.
- Directed ISO-compliant lab operations (IEC 17025 & ISO 9001), integrating LEAN management (Kaizen & 5S) to enhance workplace safety, and R&D output.
- Managed R&D proposal drafting for EU Flagship Programs (Erasmus+, Horizon) using Agile methodologies (Trello, Jira, GitLab) to synchronize milestones across +15 EMEA partners (e.g., Infenion, Imec, Siemens).
- Integrated AI-driven tools (Vertex AI, LLMs) into HW/SW workflows, automating R&D processes to accelerate development cycles.

R&D Associate Manager - Sustainable Mobility **Nov 2024 – Oct 2025**

Politecnico di Milano *Milan, Italy*

Recruited to lead a research project on electric vehicles (EV), while concurrently finalizing my Ph.D. dissertation.

- Led the design of high-precision sensor fusion systems for Sustainable Mobility (EV) applications.
- Automated test workflows by applying NI LabVIEW/TestStand, driving a 50% reduction in operational testing time for EV navigation systems compared to the legacy solution while ensuring measurement repeatability.
- Directed data-driven EV navigation predictive maintenance, employing unsupervised statistical learning and anomaly detection to ensure system reliability.

Ph.D. Researcher - Green Energy Integration & Smart Metering **Nov 2021 – May 2025**

Politecnico di Milano *Milan, Italy*

- Developed AI-driven predictive maintenance frameworks for smart metering compliance (IEC 62052/53/56)

via statistical learning, advanced signal processing, and anomaly detection with $> 90\%$ non-technical loss detection accuracy.

- Deployed scalable Cloud/IoT architectures (AWS/MQTT) for high-reliability data telemetry, ensuring 95%+ data integrity across 3+ years of continuous operation.
- Developed advanced signal processing algorithms for wearable health-tech, applying harmonic analysis to BioSignal (PPG & EEG) monitoring.
- Defined novel Signal Quality Assessment (SQA) indicators to enhance the reliability of wearable sensor data in health-tech.
- Formulated standardized metrological decision frameworks, contributing to international JCGM guidance for conformity assessment and risk-based decision frameworks.

Technical Consultant

Sep 2018 – Sep 2019

Freelance contractor [Startup ecosystems]

Ahvaz, Iran

- Deployed IIoT monitoring systems (M2M), optimizing ARM-based embedded hardware for EMC Compliance (IEEE 61000) in RF noisy environments for R&D purposes using Altium Designer & LTspice.
- Designed sensor fusion dashboards, leveraging software-driven control systems for real-time monitoring and data acquisition within the Agri-Tech sector via NI LabVIEW and Node-RED.

Co-Founder & CEO

Dec 2017 – Aug 2018

Atiyeh Hoshyar LLC. [Startup]

Ahvaz, Iran

- Co-founded an IoT startup and secured 60% of total funding through targeted pitches and incubator programs, accelerating early product development and market validation.
- Scaled engineering capacity by 40% in the first year by recruiting, onboarding, and leading a 5-person R&D team specializing in electronic design and embedded hardware/firmware.
- Filed a patent in Home Automation and Remote Monitoring, integrating secure M2M communication into ARM-based embedded HW/FW architectures.

TEACHING EXPERIENCE

Lecturer

A.Y. 2022 - A.Y. 2026 SEM I

Digital Signal Processing (Advanced) (M.Sc.)

Politecnico di Milano, Milan, Italy

- Co-delivered **196 hours** of lectures in Signal Processing, Data Acquisition Systems (DAQ), Time Synchronization & Positioning (GPS), and Instrument interfacing.
- Conducted live demonstrations, exam preparation sessions, student project evaluation, and written/oral examination rounds.

Teaching Assistant

A.Y. 2025/26 SEM I

Measurement & Instrumentation (B.Sc.)

Politecnico di Milano, Milan, Italy

- Delivered **56 hours** of hands-on lab lectures for fundamentals of electrical and electronic measurements, measurement systems for Industry 4.0, and practical demonstrations in NI LabVIEW, Python, and MATLAB.

EDUCATION

Ph.D. in Electrical Engineering (with honors)

Nov 2021 – May 2025

Politecnico di Milano

Milan, Italy

Thesis (public): Data-driven approaches for anomaly detection in energy metering.

M.Sc. in Electrical Engineering

Sep 2019 – Oct 2021

Politecnico di Milano

Milan, Italy

Thesis (public): Real-time motion capturing using inertial measurement units and applied embedded systems development

B.Sc. in Electrical Engineering

Sep 2013 – Jul 2018

Shahid Chamran University

Ahvaz, Iran

Thesis: Embedded Systems Design for Test Automation in AgriTech.

ADDITIONAL INFORMATION

Certificates	LEAN Management Fundamentals, Probability and Data with R, Python Data Structures, Data Scientist's ToolBox (Coursera); NI LabVIEW Object-Oriented Programming (Emerson); Signal and Power Integrity for High-Speed PCBs (Polimi); EU Driving License (B).
Awards	IEEE AMPS 2023 - Best Presentation Award; MIUR Scholarship for PhD Studies.
Speaking	Invited speaker at Milano Digital Week 2024, organized by the National Order of Engineers in Italy (CNI) for a 3-year regional energy monitoring project.
Committees	Technical Program Committee (TPC) for IEEE MetroXRaine (since 2024).
Memberships	GMEE (Gruppo Misure Elettriche ed Eletttroniche); IEEE Instrumentation & Measurement Society & IEEE Young Professionals (since 2021).
Languages	English (C2); Italian (B1).
Visa Sponsorship	Not required.

PUBLICATIONS

(J): Journal Publications (C): Conference proceedings

- (J) Risk-aware decisions: Taking into account admissible risk and measurement uncertainty in setting the acceptance limits - The European Physical Journal Conferences 2025
- (J) Obtaining a Probability Distribution From a Unimodal Possibility Distribution - IEEE Transactions on Instrumentation and Measurement 2024
- (J) A general Monte-Carlo approach to consider a maximum admissible risk in decision-making procedures based on measurement results - ACTA IMEKO 2023
- (J) Deconvolution-based methods to extract uncertainty components - Measurement Science and Technology 2023
- (J) A method to consider a maximum admissible risk in decision-making procedures based on measurement results - ACTA IMEKO 2023
- (J) Low-cost real-time motion capturing system using inertial measurement units - ACTA IMEKO 2022
- (C) Enhanced Resolution Analysis of Photoplethysmography Signals for Stress Monitoring Using Taylor-Fourier Method - IEEE EMBC 2025
- (C) Photoplethysmography Signal Quality Assessment Using Instantaneous Harmonic Analysis via Taylor-Fourier Method - IEEE I2MTC 2025
- (C) State of Charge Estimation of an Electric Battery using the Theory of Evidence - IEEE I2MTC 2025
- (C) Data-Driven Anomaly Detection for Smart Energy Meters - IEEE AMPS 2024
- (C) A Method to Obtain a Probability Distribution from a Unimodal Possibility Distribution - IEEE I2MTC 2024
- (C) Novel Algorithms for Filtering and Event Detection in Non-Intrusive Load Monitoring - IEEE AMPS 2024